

(Cyber worked) in IoT-Based Remote Patient Monitoring System for Continuous Home Healthcare

My responsibility is to **protect patient data** from the sensor → cloud → dashboard.

STEP BY STEP IMPLEMENTED IN MY WORK :

1 . Threat Model (First Cyber Step) :

Risk	Example
Data interception	Hacker reads heart rate
Unauthorized access	Fake doctor login
Data tampering	Wrong vitals sent
Privacy violation	Patient data leaked

I mention this in **project documentation**.

2 . Secure Data Transmission

- ESP32 sends data over Wi-Fi → can be intercepted.
- I using in (**Encrypted communication**)

IT'S IMPLEMENTED 2 OPTION OR USING IN THIS :

Option A: HTTPS (Recommended for Projects)

ESP32 → Cloud using HTTPS (TLS encryption)

- ✓ Data encrypted
- ✓ Easy to explain
- ✓ Widely accepted

Option B: MQTT over TLS (Advanced)

ESP32 → Broker → Dashboard

- ✓ Strong encryption
- ✓ Used in hospitals

3 . Data Encryption (AES-256)

Even before sending data, you **encrypt the payload**.

THAT MEAN : Patient vital data is encrypted using AES-256 before cloud transmission.

4 . Authentication & Access Control

It's access only Doctor Authorized caregiver

- **API keys and authentication** are used to prevent unauthorized access.
- Cloud dashboard access is limited to **authorized doctors/caregivers only**.

5 . Secure Cloud Configuration :(If using ThingSpeak / Firebase:)

No personal identifiers are stored to ensure patient privacy.

Security Steps:

- Private channels
- API key protection
- HTTPS enabled
- Read/write permissions separated

THIS UNDERSTAND EASILY

ThingSpeak private channels ensure only authorized clients can read patient data.

6 . Data Privacy & Compliance (HIPAA / GDPR)

Data integrity is maintained to prevent **tampering or false readings**.

No personal identifiers are stored to ensure **patient privacy**.

The system follows basic **HIPAA/GDPR security principles**.

THAT MEANS :

The system complies with HIPAA/GDPR guidelines by encrypting patient data and limiting access to authorized users only.

OVERALL THE THE MY WORK IMPLEMENTED IN THIS :

Sensor → ESP32

- Encrypted Data (AES)
- Secure WiFi (HTTPS)
- Cloud Server
- Authenticated Dashboard

So I work in I ensured secure transmission, encryption, and privacy of patient health data in the IoT-based monitoring system.